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# Does the gender equity mismatch explain Korea's fertility collapse? Evidence from Korea and the OECD

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**Korea's total fertility rate reached 0.72 in 2023, the lowest ever recorded for a country. The most cited explanation is the gender equity mismatch: women gained near equal access to education and jobs while the home stayed traditional. This paper tests that explanation two ways. Within Korea, fertility and every equity measure trend steadily over the same decades, so a single country's time series cannot distinguish the mismatch from any other explanation. Across 30 OECD countries, an absolute mismatch index relates to fertility in the predicted direction but weakly ($r = -0.25$), and half of that comes from Korea alone. The share of births outside marriage predicts fertility far better: countries where birth is tied to marriage average 1.15 births per woman against 1.39 elsewhere, and once it enters the model the mismatch coefficient drops to zero. Even so, Korea sits 0.36 births below what the model predicts, and the model explains only about a quarter of the variation across countries. The mismatch is real in Korea, but in this data it is not what separates Korea from its peers, and neither variable captures most of what separates very low fertility from ordinary fertility among rich countries.**

fertility | gender equity | Korea | OECD | marriage

## Significance

Fertility is falling across the developed world. As the total fertility rate (TFR) drops further below the replacement rate of 2.1 births per woman, societies will struggle to sustain themselves. New generations are the foundation of social progress and are needed to sustain the institutions, economies and cultures that earlier generations built. Understanding why fertility falls so far in some wealthy countries is therefore an aim of utmost importance. Korea, where the TFR has fallen to 0.72, is the extreme case. This paper tests the leading explanation, a mismatch between women's equality in education and work and inequality at home, against Korea's own history and against 30 OECD countries, and asks what else could account for the decline.

## Introduction

For decades the standard explanation for falling fertility was economic. As women gained career opportunities, raising children carried a higher opportunity cost, so people had fewer of them. Doepke and coauthors (1) show this no longer holds in rich countries. The consensus now is that fertility is tied to the compatibility of family and career life. McDonald (2) hypothesizes that fertility collapses where women have nearly equal access to education and work but the division of labor at home stays traditional, and argues that the countries with the lowest fertility exhibit this mismatch. Goldin (3) adds that these countries modernized faster than their gender norms could adjust.

South Korea is the extreme case. Its total fertility rate (TFR) reached 0.72 in 2023, the lowest ever recorded for a country, despite two decades of heavy spending on family policy (4). This paper uses Korea as a test of the leading explanation for very low fertility. If that explanation is right anywhere, it should be right in Korea. The goal is to see whether the current consensus holds up in the country where the decline has gone furthest, and if it does not, to identify what other factors are at play.

The answer, in this data, is mostly no. Within Korea, the time series cannot tell the hypothesis apart from any other explanation. Across the OECD, the mismatch has a weak relationship with fertility that disappears once one accounts for how tightly births are tied to marriage. Korea then remains an outlier that neither variable explains.

**Related work.** McDonald (2) proposed that very low fertility occurs when equity in individual oriented institutions (education, employment) runs ahead of equity in family oriented institutions. Esping-Andersen and Billari (5) extended this to a U shape in which fertility recovers once the private sphere catches up, citing the Nordic countries. Goldin (3) applies the same logic to countries that grew rich quickly, arguing that norms in the home lag behind the economy. On Korea specifically, Anderson and Kohler (6) emphasise the cost of private education, and Raymo and coauthors (7) emphasise the retreat from marriage across East Asia, where births outside marriage are rare. Lesthaeghe (8) describes a broader shift in family norms, including cohabitation and childbearing outside marriage, that spread unevenly across rich countries.

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This paper extends that work in two ways. First, it builds an absolute mismatch index, expressed in percent of gender parity on both sides, so that countries can be compared on the gap the theory names rather than on trends. Second, it tests the marriage channel directly by adding the share of births outside marriage and its interaction with the mismatch.

## Data

**Sources and time window.** Table 1 lists the sources. Fertility, labor force participation, tertiary enrollment, and GDP per capita are pulled from the World Bank API (9). Men's and women's minutes of unpaid work come from the OECD Time Use Database (10). The share of births outside marriage comes from the OECD Family Database table SF2.4 (11). For Korea alone, men's share of unpaid work every five years from 1999 to 2024 is taken from the Statistics Korea Time Use Survey via KOSIS (12), and fathers' share of parental leave takers from Ministry of Employment and Labor press releases (13).

**Table 1. Data sources. Parity measures are scaled so that 100 means men and women are equal.**

| Sphere | Measure | Coverage | Source |
|---|---|---|---|
| Outcome | Total fertility rate | Korea 1980–2024; OECD latest (2024) | World Bank |
| Public | Female/male labor force participation, % | Korea 1990–2025; OECD latest (2024) | World Bank / ILO |
| Public | Female/male tertiary enrollment, % | Korea 2000–2024; OECD latest (2019–2024) | World Bank / UNESCO |
| Private | Men's share of unpaid work, % | Korea 1999–2024 (every 5 years); OECD one survey per country | KOSIS; OECD Time Use |
| Private | Fathers' share of parental leave takers, % | Korea 2010–2024 | MOEL |
| Marriage | Births outside marriage, % of births | OECD latest (2018–2024) | OECD Family Database |
| Control | GDP per capita, PPP, constant $ | OECD latest | World Bank |

**Unit of analysis.** For Korea, each row is a year, 1980 to 2024. For the OECD, each row is a country, using the latest value of each indicator. All data are national aggregates; no individual level data are used.

**Analysis sample.** Table 2 characterises the 30 countries in the OECD analysis. All 38 members have World Bank data. The inner merge with the time use data drops eight members without a diary survey in the OECD database (Chile, Colombia, Costa Rica, Czechia, Iceland, Israel, Slovakia, Switzerland). The merge with the marriage data keeps all 30.

**Limitations and missingness.** The OECD time use data is one survey per country, and the API does not return the survey year (roughly 2009 to 2024), so the age of each value is unknown. The marriage measure is defined slightly differently for Australia, Japan, Korea and New Zealand. Most marriage values are for 2023; Ireland is 2019, Belgium 2018 and Denmark 2021. The Korean housework and leave series are short (six and ten points) and were typed in from KOSIS tables and press releases. TFR is a period measure and is pushed down when births are delayed, so the level overstates the decline, though cohort fertility in Korea is also falling.

## Methods

The analysis went in three steps. Each step was a response to what the one before it could not do.

**Step 1. Korea alone.** Korea is the extreme case, so the first question is whether the mismatch shows up in Korea's own history. The Korean series are joined on year (left join onto 1980 to 2024, leaving 45 rows with fertility). I first compute the correlation between TFR and the raw values of each equity measure, year by year. Every series moves in one direction over the same decades, so paired with TFR, every series will produce a correlation near $\pm 1$ whether or not they are related. The real check uses changes from one year to the next (five year changes for the time use survey), which removes the shared trend. Because these factors might affect fertility with a delay, I also shift each series forward by one to four years and check whether the correlation of changes becomes both larger than chance would produce and negative, as the theory predicts, at any offset. This is done only for female labor force participation and tertiary enrollment. The housework and leave series have 5 to 7 changes which is too few to scan without finding patterns by chance. Last, I build a mismatch score within Korea by standardizing each series, averaging the two public and the two private measures, and subtracting private from public. Code is in `01_korea_timeseries.ipynb`..

Step 1 shows that Korea's time series cannot test the hypothesis. Everything trends together, and a within country score measures how fast each sphere is changing, not how far apart they are. That is why step 2 brings in the rest of the OECD.

**Step 2. The OECD.** The 38 OECD members are the comparison set, so that income, health systems and access to contraception are roughly held constant. World Bank data for all 38 are inner merged with the time use table on country code (38 to 30 rows).

Each sphere needs one measure on the same scale. For the public sphere I average two ratios, female to male labor force participation and female to male tertiary enrollment, both scaled so that 100 represents equality. Education and paid work are the two institutions McDonald names, and both ratios exist for every member. For the private sphere I use men's share of unpaid work from time diaries, divided by 50 so that 100 is again equality. This is the most

**Table 2. The 30 OECD countries in the cross section, sorted by mismatch. Public parity averages the labor force and tertiary ratios; private parity is men's unpaid work share divided by 50; mismatch is public minus private. TFR is for 2024. Marriage year is the year of the births outside marriage value.**

| Country | TFR | Public parity | Men's unpaid share (%) | Private parity | Mismatch | Births outside marriage (%) |
|---|---|---|---|---|---|---|
| Portugal | 1.40 | 102.8 | 22.7 | 45.4 | 57.5 | 59.5 |
| Japan | 1.20 | 87.8 | 18.4 | 36.9 | 51.0 | 2.4 |
| Korea | 0.75 | 84.0 | 18.6 | 37.1 | 46.9 | 4.7 |
| New Zealand | 1.60 | 113.3 | 34.8 | 69.6 | 43.7 | 48.4 |
| Türkiye | 1.50 | 79.5 | 18.1 | 36.3 | 43.2 | 3.1 |
| Ireland | 1.50 | 101.9 | 30.3 | 60.5 | 41.4 | 38.4 |
| Latvia | 1.20 | 107.6 | 33.9 | 67.7 | 39.8 | 37.3 |
| Italy | 1.20 | 99.6 | 29.9 | 59.8 | 39.7 | 40.5 |
| Lithuania | 1.10 | 107.4 | 34.2 | 68.4 | 39.1 | 27.3 |
| Australia | 1.50 | 109.7 | 35.6 | 71.1 | 38.6 | 39.9 |
| Spain | 1.10 | 104.0 | 33.5 | 67.1 | 36.9 | 50.0 |
| Greece | 1.20 | 90.7 | 27.2 | 54.3 | 36.3 | 9.7 |
| United Kingdom | 1.60 | 107.4 | 36.0 | 72.1 | 35.3 | 47.6 |
| Slovenia | 1.50 | 108.8 | 36.8 | 73.5 | 35.2 | 56.5 |
| Luxembourg | 1.20 | 101.8 | 33.6 | 67.1 | 34.7 | 39.0 |
| Estonia | 1.20 | 110.7 | 39.9 | 79.9 | 30.8 | 53.8 |
| France | 1.60 | 105.6 | 37.6 | 75.2 | 30.5 | 58.5 |
| Belgium | 1.40 | 105.9 | 37.8 | 75.6 | 30.3 | 52.4 |
| Austria | 1.30 | 103.8 | 36.8 | 73.6 | 30.2 | 40.0 |
| Hungary | 1.40 | 100.8 | 35.6 | 71.2 | 29.6 | 24.4 |
| Mexico | 1.90 | 88.8 | 29.7 | 59.4 | 29.3 | 73.7 |
| Poland | 1.10 | 107.6 | 39.3 | 78.6 | 29.0 | 28.7 |
| Sweden | 1.40 | 115.5 | 43.7 | 87.4 | 28.1 | 57.4 |
| Canada | 1.20 | 107.1 | 39.9 | 79.7 | 27.4 | 29.0 |
| United States | 1.60 | 108.0 | 40.4 | 80.8 | 27.2 | 40.0 |
| Norway | 1.40 | 111.8 | 43.0 | 86.0 | 25.8 | 61.2 |
| Netherlands | 1.40 | 101.8 | 39.3 | 78.5 | 23.3 | 42.1 |
| Denmark | 1.50 | 108.7 | 43.4 | 86.8 | 21.9 | 54.7 |
| Finland | 1.20 | 107.2 | 43.7 | 87.4 | 19.8 | 48.4 |
| Germany | 1.40 | 96.8 | 39.5 | 78.9 | 17.9 | 33.1 |
| Mean | 1.35 | 102.9 | 34.4 | 68.9 | 34.0 | 40.1 |

direct proxy for equality at home that I could find. Mismatch is public minus private, so it represents the difference in equality within the two spheres.

The mismatch is tested by its correlation with TFR, by splitting countries into three equal groups by mismatch and comparing their average fertility, by checking whether the correlation survives once differences in income are taken out of both variables, and by dropping Korea, Mexico and Türkiye to see whether any single country drives the result.Code is in `02_oecd_mismatch.ipynb`.

Step 2 finds a weak relationship, half of it from Korea, with several high mismatch countries at ordinary fertility. That raised a question about mechanism. In Korea almost every birth happens inside marriage, so an unequal home is a reason not to marry, and not marrying means no children. In France or Sweden a couple can have children without marrying. If that matters, the mismatch should hurt fertility more where births require marriage. Step 3 tests this.

**Step 3. Does marriage matter?.** The share of births outside marriage (OECD Family Database SF2.4) is merged onto the 30 countries (inner join, 30 to 30). I check it on its own, then split countries at its median and compare the mismatch correlation in each half. The main test is four regressions, each adding one term: TFR on mismatch; adding births outside marriage; adding their interaction; adding log GDP per capita. I subtract the average from each variable before multiplying them, so the mismatch coefficient represents its effect in a country with a typical share of births outside marriage. The theory predicts a positive interaction. For each country I also check how much the fitted line moves if that country is removed, to see whether any single country drives the result. Code is in `03_marriage_regression.ipynb`.

## Results

**Within Korea, the time series cannot test the hypothesis.** Figure 1 shows the raw series. Fertility fell in two main steps, a steep drop to about 1.6 by the mid 1980s and a slower slide to 0.72. The equity series begin only in 1990 to 1999, so they can speak only to the second part. Over that period both spheres improved. Female labor force participation reached 57 percent by 2024, female tertiary enrollment passed 100 percent of the age group, men's share of unpaid work rose from 10.6 percent (1999) to 22.4 percent (2024), and fathers' share of leave takers rose from 2.7 percent (2010) to 31.6 percent (2024). Women still do 3.5 times the housework. The mismatch clearly exists.

Table 3 shows why that is not evidence. In levels, every equity measure correlates with TFR at −0.7 to −0.99. Taken literally these say that equalization lowers fertility, which is the first sign that the correlations only reflect shared

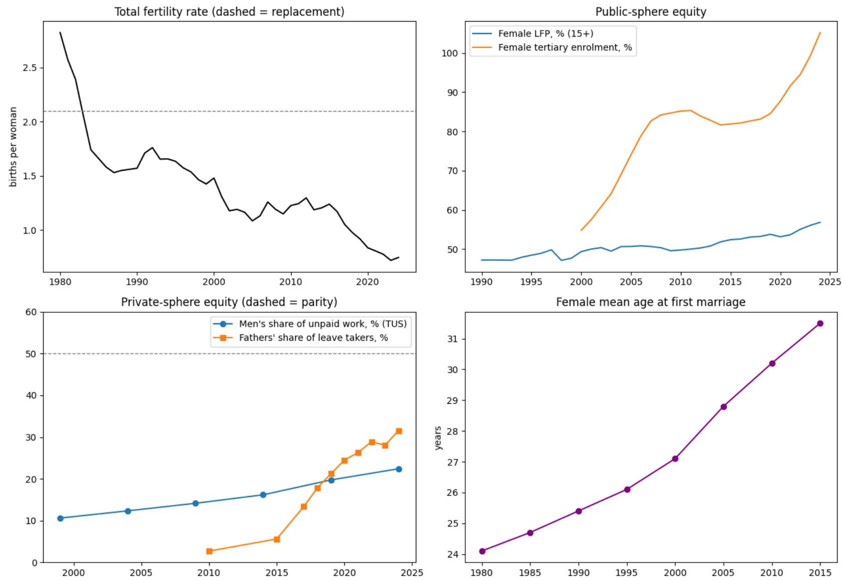


**Fig. 1.** Korea, 1980 to 2024. Top left: total fertility rate (dashed line is replacement). Top right: public sphere measures. Bottom left: private sphere measures (dashed line is parity). Bottom right: female mean age at first marriage. Every equity measure rises while fertility falls.

trends. In first differences they collapse to 0.07, 0.01, 0.12 and 0.01. Short run movements in equity have no detectable relationship with short run movements in fertility. Allowing for a delay does not help. labor force participation shows nothing at any offset. Tertiary enrollment reaches 0.47 at a four year offset, but the sign is the opposite of what the theory predicts (rising enrollment goes with rising fertility), and it peaks at the last offset checked, which suggests that some trend is still left in the data rather than a real delayed effect.

**Table 3. Korea: correlation between TFR and each equity measure, using raw values and then using year to year changes. The raw value correlations are high only because every series trends over the same decades.**

| Measure | $r$, levels | years | $r$, changes | changes |
|---|---|---|---|---|
| Female labor force participation | $-0.914$ | 35 | $0.068$ | 34 |
| Female tertiary enrollment | $-0.725$ | 25 | $0.012$ | 24 |
| Male minus female participation gap | $0.951$ | 35 | $0.119$ | 34 |
| Fathers' share of leave takers | $-0.989$ | 10 | $0.012$ | 7 |
| Men's share of unpaid work (5 yearly) | $-0.939$ | 6 | $-0.527$ | 5 |

The composite mismatch score makes the problem concrete. It correlates with TFR at $+0.40$ over 2000 to 2024, the opposite of the prediction. After 2015 the private sphere score outpaced the public one, driven almost entirely by the surge in paternal leave, which was itself a policy response to low fertility. This suggests that a standardized, within country mismatch measures speed of change along each sphere's own path, but not the absolute gap the theory is about. Each standardized value only says how far that year sits from the series' own average. It does not say how far the series is from true equality. And fathers' leave rose because the government expanded it in response to low fertility, so it is partly caused by low fertility and cannot be used to explain it. The absolute gap can only be tested across countries.

**Across the OECD, the mismatch relates to fertility weakly, and mostly through Korea.** On the public side Korea is below most peers. Its participation ratio (78.9) is in the lower group and its tertiary ratio (89.1) is the lowest in the OECD. On the private side Korea sits at the bottom with Japan and Türkiye, with men doing about 18 percent of unpaid work against 43 to 44 percent in the Nordic countries. Korea's mismatch score of 46.9 ranks third after Portugal (57.5) and Japan (51.0).

Figure 2 shows each component against fertility. Neither public measure has a clear relationship with TFR. Men's share of unpaid work is positively related ($r = 0.27$). Countries where men do more at home have somewhat higher fertility, which makes sense. The relationship is loose, however. Finland, Germany and Poland have fairly equal households and low fertility; Ireland and New Zealand have unequal households and ordinary fertility.

Figure 3 is the main test of the composite. The correlation is $-0.25$ across 30 countries, the direction the theory predicts. Terciles of mismatch average 1.28, 1.36 and 1.43 births from high to low, a gradient of 0.15 births across the whole range. The lowest fertility cluster (Korea, Japan, Italy, Spain, Lithuania, Latvia) sits on the high mismatch side, but so do Portugal, Ireland and New Zealand with ordinary fertility. Within the OECD, income is unrelated to fertility ($r = -0.07$), and removing it leaves the mismatch and housework correlations unchanged ($-0.27$ and 0.31). Dropping Mexico and Türkiye changes nothing ($-0.27$). Dropping Korea alone cuts the correlation in half, to $-0.13$. Half of the OECD wide pattern is one country.

**Marriage absorbs the mismatch.** Korea (4.7 percent), Japan (2.4), Türkiye (3.1) and Greece (9.7) are the only OECD countries where fewer than 10 percent of births occur outside marriage; the median is 40 percent and the Nordic

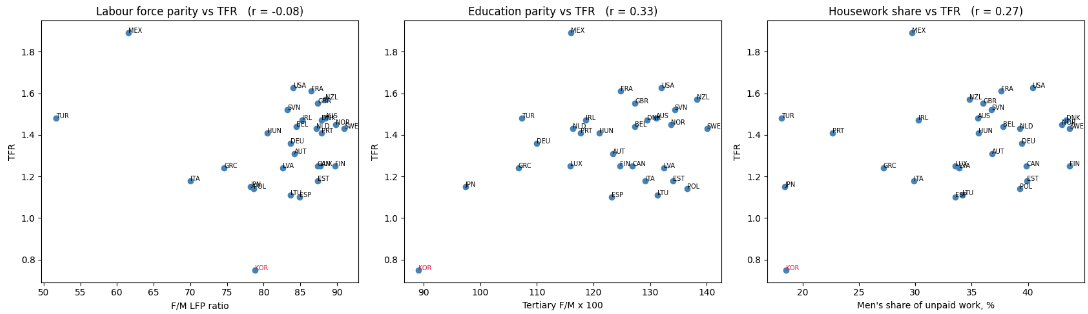


**Fig. 2.** Each component of the index against TFR across 30 OECD countries. Public measures show no relationship; men's share of unpaid work is weakly positive.

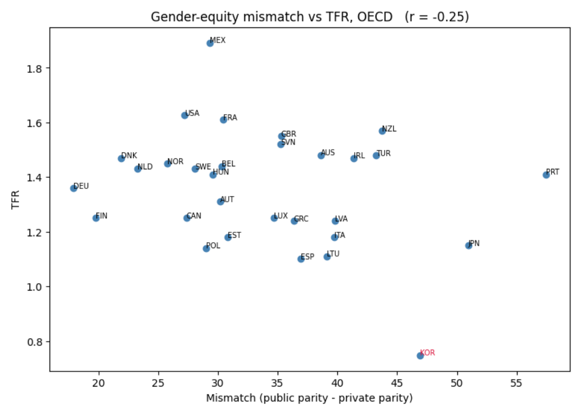


**Fig. 3.** Mismatch (public parity minus private parity) against TFR across 30 OECD countries, $r = -0.25$. Without Korea, $r = -0.13$.

countries, France, Portugal and Mexico exceed 50 percent (Figure 4). This share is positively related to fertility at $r = 0.57$, twice the size of the mismatch relationship, and it is not Mexico and Korea pulling the line: without both, $r = 0.36$. The share is also correlated with the mismatch ($r = -0.34$).

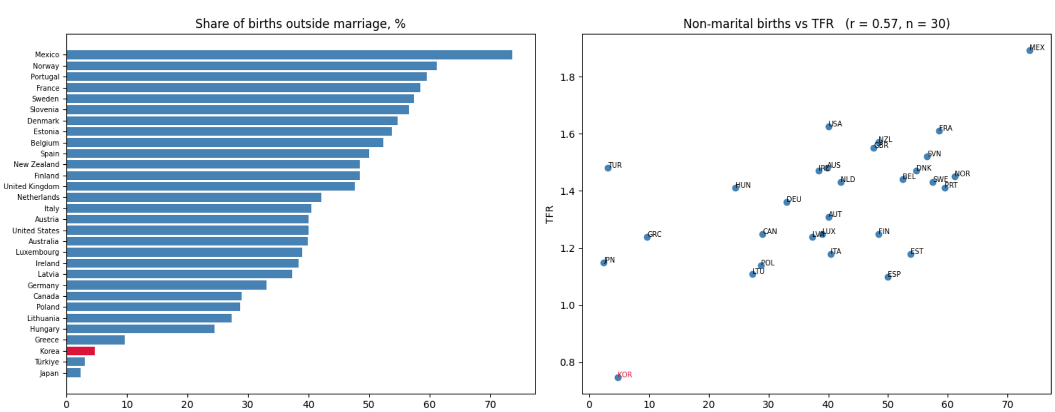


**Fig. 4.** Left: share of births outside marriage across the OECD, Korea in red. Right: the same share against TFR, $r = 0.57$.

The simplest version of the interaction test splits countries at the median share. The mismatch correlation is $-0.38$ in the half where births mostly require marriage and $-0.05$ in the other half. That looks like the predicted pattern, but the first half contains Korea, and without Korea it is $-0.18$.

Table 4 reports the regressions. Model 1 uses the mismatch alone and gives the same picture as the correlation in step 2. The coefficient is $-0.006$ births per point of mismatch, the confidence interval includes zero, and adjusted $R^2$ is 0.03. Model 2 adds the share of births outside marriage and is the important one. Adjusted $R^2$ rises to 0.28. The coefficient on births outside marriage is $+0.0066$ per percentage point ($p = 0.003$), which works out to about 0.33 births between a country at 5 percent and one at 55 percent. The mismatch coefficient falls to $-0.0015$ ($p = 0.70$). Model 3 adds the interaction, which is where the theory predicted an effect, and the estimate is 0.0000 ($p = 0.90$). This is not an effect that is present but too noisy to detect; the estimate itself is zero. Model 4 adds income and changes nothing.

Without Korea the marriage coefficient shrinks slightly ($+0.0052$, $p = 0.013$) but keeps its sign, while the mismatch coefficient is indistinguishable from zero ($-0.0001$). Cook's distance identifies Korea (0.47), Türkiye (0.32) and Mexico (0.28) as the influential countries; the next largest is 0.05. Türkiye and Mexico matter because they combine extreme marriage shares with the two highest fertility rates. Korea matters because it is far below where the model puts it.

**Table 4. OLS regressions of TFR on the mismatch and the share of births outside marriage, 30 OECD countries (models 1 to 4) and 29 without Korea. Predictors are centered on their means. 95% confidence intervals in brackets.**

| | All countries ($n = 30$) | | | | Without Korea ($n = 29$) | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (2) | (3) |
| Mismatch | $-0.0059$ | $-0.0015$ | $-0.0015$ | $-0.0021$ | $-0.0001$ | $0.0003$ |
| | $[-0.015, 0.003]$ | $[-0.010, 0.007]$ | $[-0.010, 0.007]$ | $[-0.011, 0.006]$ | $[-0.008, 0.008]$ | $[-0.007, 0.008]$ |
| Births outside marriage | | $0.0066$ | $0.0067$ | $0.0067$ | $0.0052$ | $0.0059$ |
| | | $[0.003, 0.011]$ | $[0.002, 0.011]$ | $[0.002, 0.011]$ | $[0.001, 0.009]$ | $[0.002, 0.010]$ |
| Mismatch × marriage | | | $0.0000$ | $0.0000$ | | $-0.0002$ |
| | | | $[-0.0005, 0.0004]$ | $[-0.0005, 0.0004]$ | | $[-0.0006, 0.0002]$ |
| log GDP per capita | | | | $-0.072$ | | |
| | | | | $[-0.277, 0.133]$ | | |
| Intercept | 1.358 | 1.358 | 1.357 | 1.357 | 1.373 | 1.365 |
| Adjusted $R^2$ | 0.029 | 0.283 | 0.256 | 0.242 | 0.169 | 0.162 |

**Korea remains unexplained..** With both variables in the model, Korea is predicted at about 1.11 births and sits at 0.75, a residual of $-0.36$, by far the largest in the sample. It is also the lowest of the four countries where fewer than 10 percent of births occur outside marriage: Japan and Greece stand at 1.20 and Türkiye at 1.50. The next largest discrepancies are Spain ($-0.32$), Estonia ($-0.28$) and Finland ($-0.19$). In all three, births outside marriage are common, so the countries the model fails to explain are not only those where childbearing requires marriage. The full model explains only about a quarter of the variation across countries.

## Discussion

This paper asked whether the gender equity mismatch, the most cited explanation for Korea's fertility collapse, is visible in the data. The answer came in three parts.

First, Korea's own history cannot answer the question. Every equity measure rose over the same decades that fertility fell, so each correlates with fertility near $-1$; but any two trending series will do so regardless of any real connection. Once the shared trend is removed through first differences, the relationships vanish. The mismatch index even moves in the wrong direction, driven by the post 2015 surge in paternal leave, which was itself a policy response to low fertility and is therefore partly endogenous to the outcome. The conclusion is not that the mismatch does nothing in Korea, but that a single country's time series cannot distinguish it from any rival explanation.

Second, in the cross national comparison the theory thus calls for, the mismatch performs worse than its prominence suggests. Countries with a larger gap between public and private equality have somewhat lower fertility, in the direction McDonald (2) predicts, but the relationship is weak, half of it comes from Korea alone, and Portugal, Ireland and New Zealand combine large mismatches with ordinary fertility.

Third, a different variable does most of the work: whether childbearing is institutionally tied to marriage. In the four countries where nearly all births occur inside marriage (Korea, Japan, Türkiye, Greece), fertility averages 1.15 births per woman against 1.39 elsewhere. Once this share enters the model, the mismatch coefficient collapses to zero, and the interaction the theory predicts, that an unequal home should be more costly where children require marriage, is estimated at exactly zero.

None of this directly refutes the hypothesis. Korea does have the configuration the theory describes, and the private sphere is where it differs most from its peers. The mismatch could still operate through marriage itself, if an unequal home deters marriage and unmarried couples forgo children. But if so, the mismatch should predict fertility within the marriage tied countries, and it does not. The marriage variable is also open to interpretation: a high share of nonmarital births signals that marriage is optional for family formation, but it equally tracks the broader liberalization of family norms that demographers link to fertility recovery in the Nordic and French cases (8). It is safer to read it as marking the institutional coupling of birth and marriage than as a cause.

The most interesting result may be the residual. With both variables in the model, Korea sits 0.36 births below its predicted value, and it is by far the lowest of the marriage tied countries: Japan and Greece stand at 1.20 and Türkiye at 1.50, against Korea's 0.75. The leading explanation does not separate Korea from its peers. The shortfall isn't just Korea's; the full model explains just over a quarter of the cross national variation, and Poland, Finland and Mexico sit far from the regression line too. Perhaps something particular to Korea, whether the collapse of marriage itself, working hours, housing or private education costs (6), accounts for some of Korea's gap, though even if it did, it would not explain the broader discrepancy between the consensus hypothesis and what the data show across the OECD as a whole.

Limitations come from the data: thirty countries, three influential points (Korea, Türkiye, Mexico), and time use snapshots of unknown and varying age. Future work should use time use surveys from matched years, add a behavioral private sphere measure such as mothers' employment after first birth, and test how marriage affects things within East Asia, where marriage timing varies widely even though nonmarital births are uniformly rare.

**ACKNOWLEDGMENTS.** Code, data and figures are at https://github.com/clewner/fertility-decline. The project website is at https://qssfertility.vercel.app.

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## Agentic Analysis

**How the assistant was used.** I used Claude throughout the project. Early on it was most useful for orienting myself in the subject and the existing literature. Later it did most of the heavy lifting on data analysis, and I used it on all three notebooks. It also generated the first draft of the project website.

**What I asked.** Background reading on the gender equity mismatch and related work; help pulling and merging the data; coding for each notebook; sanity checks on results; and standard practice questions about how to approach particular sub-analyses. For the website, I asked it to read the paper and notebooks and generate a minimalist site from them.

**What I accepted.** The one statistical tool in this paper that I did not already know is Cook's distance. I asked whether there was a standard way to quantify how much a single country moves a regression, and Claude suggested it. More generally I accepted most of its coding, which saved a great deal of time, and I bounced many ideas off it. It was good at catching mistakes and at suggesting how to structure a sub-analysis once I had decided what the question was.

**What I rejected or changed.** The recurring problem was that Claude would propose an approach that was not very good, present it confidently, and then build layers of complexity on top that made the weakness harder to see. For example, it wanted to use raw, unnormalized values rather than z scores or percentages. It wanted the time use survey to be the primary private sphere measure, even though that series has very few observations, and I judged fathers' share of parental leave takers to be the better choice. And it wanted to correlate each factor with fertility in levels, which produced very high correlations; only after I decided to look at year to year changes did the real relationship, or lack of one, become visible. On the website, the draft it produced had a lot of filler text and decorative lines that I removed by hand.

**Where the assistant went wrong.** The main errors were analytical rather than technical. It defaulted to level correlations on trending series and to unnormalized measures, and it tended to add complexity where a simpler check was needed. The website draft was also way overdone.

**Transcripts.** https://github.com/clewner/fertility-decline/tree/main/transcripts